

AI WHATSAPP CHATBOT AUTOMATION USING N8N AND GPT-4.0 MINI

:Seamless Conversational AI over WhatsApp

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Introduction

- This project automates conversations on WhatsApp by integrating an AI chatbot using **n8n**, **GPT-4.0 Mini**, and **WhatsApp Cloud API**.

The bot can interpret both **text** and **image inputs** and respond accordingly. It serves as a 24/7 assistant for **customer support**, **education**, or **personal tasks**.

Key Benefits:

- Instant replies powered by GPT-4.0 mini
- Cost-effective and fast inference
- Fully automated and scalable system



What the Bot Can Do

- **Text Questions:** Answers factual, conversational, or task-based queries

- **Image Interpretation:** Generates captions or analyzes uploaded photos

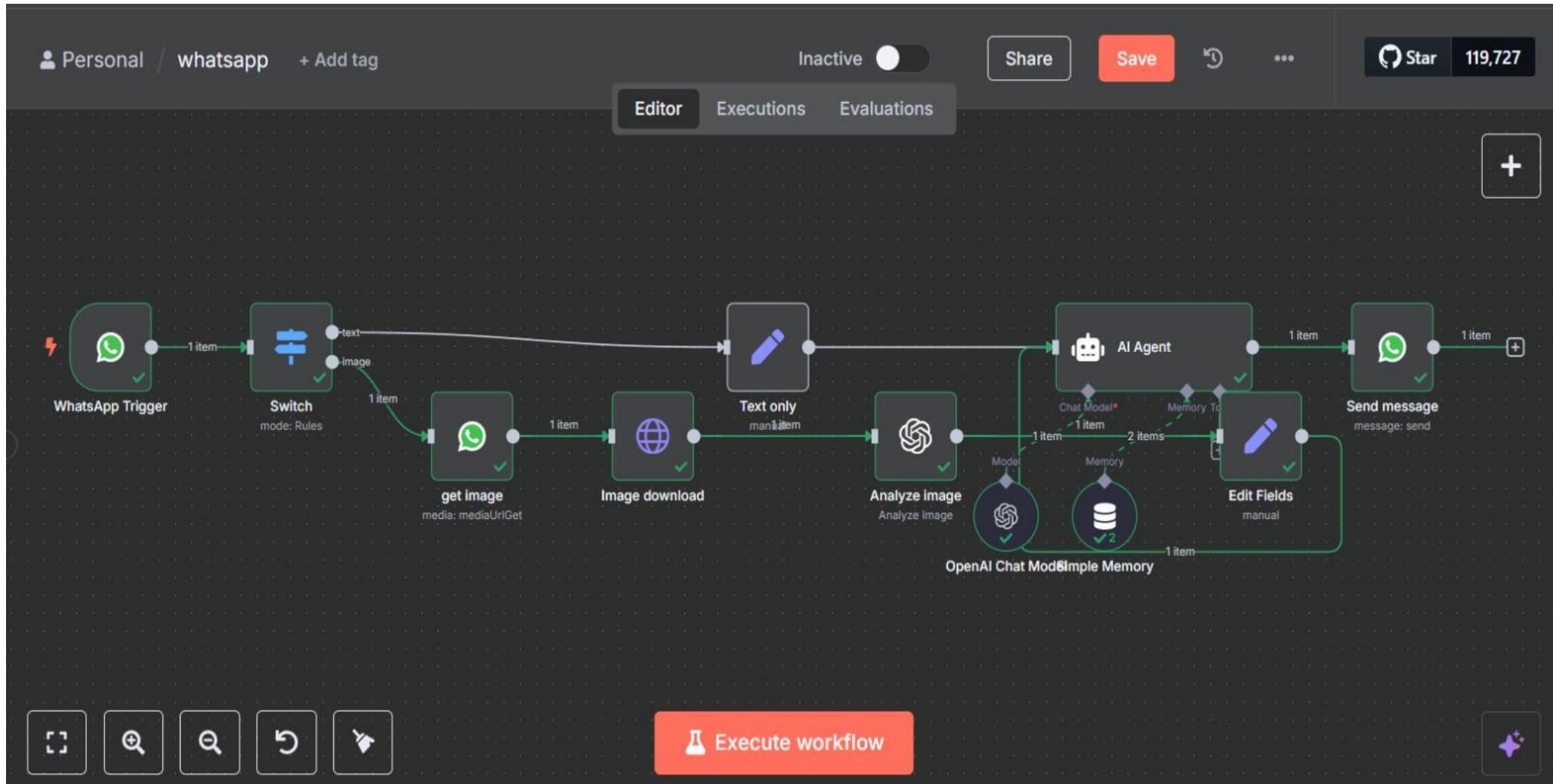
- **Smart Replies:** Remembers previous prompts in a session

- **Multi-use:** Customer support, helpdesk, feedback system

- **Hands-free access:** Just chat on WhatsApp, no app needed



Workflow



Step 1 : SET UP whatsapp reciever

- Configured using **WhatsApp Business Cloud API**:
- Registered app on Facebook for Developers
- Webhook set to receive messages and media
- Access token and phone number ID integrated into n8n
- Triggers whenever a new message is received

Parameters

Settings

Docs 

> Webhook URLs

Credential to connect with

whatsapp helperAI



Due to Facebook API limitations, you can use just one WhatsApp trigger for each Facebook App

Trigger On

Messages 



Options

No properties

Add option

Step 2 – Switch

- This node checks if the incoming WhatsApp message contains:
- Only **text** (→ go to GPT directly)
- An **image** (→ go to image download step)

This conditional logic ensures the flow follows the correct path.

The screenshot shows the configuration for a 'Switch' node in a workflow editor. The node is titled 'Switch' and has tabs for 'Parameters', 'Settings', and 'Docs'. The 'Parameters' tab is active.

Mode: A dropdown menu is set to 'Rules'.

Routing Rules:

- Rule 1:** The expression `{{ $json.messages[0].text }}` is entered in the 'fx' field. Below it, the 'exists' checkbox is checked. To the right, there is a 'Rename Output' toggle (turned on) and an 'Output Name' field containing 'text'.
- Rule 2:** The expression `{{ $json.messages[0].image }}` is entered in the 'fx' field. Below it, the 'exists' checkbox is checked. To the right, there is a 'Rename Output' toggle (turned on) and an 'Output Name' field containing 'image'.

Add Routing Rule: A button to add a new routing rule.

Convert types where required: A toggle switch is currently turned off.

Options: A section labeled 'No properties'.

OUTPUT Of Switch

OUTPUT  



Schema

Table

JSON

Text (1 item)

Image

```
[
  {
    "messaging_product": "whatsapp",
    "metadata": {
      "display_phone_number": "15551419055",
      "phone_number_id": "733403149853488"
    },
    "contacts": [
      {
        "profile": {
          "name": "Mayank"
        },
        "wa_id": "18144188274"
      }
    ],
    "messages": [
      {
        "from": "18144188274",
        "id": "wamid.HBgLMTgxNDQxODgyNzQVAgASGBYzRUlwRTU5N0E4OTgQTjGMTlGQzhCAA==",
        "timestamp": "1752925736",
        "text": {
          "body": "hello"
        },
        "type": "text"
      }
    ],
    "field": "messages"
  }
]
```

TEXT ONLY

Once a text message is received:

- Message content is extracted
- Sent to OpenAI GPT-4.0 Mini via HTTP Request node
- AI generates a smart, relevant response
- Message is sent back via WhatsApp
Quick, simple, and intelligent.

 Text only

Execute step

Parameters

Settings

Docs

Mode

Manual Mapping

Fields to Set

text

A String

=

{{ \$json.messages[0].text.body }}

hello

Drag input fields here or Add Field

Include Other Input Fields

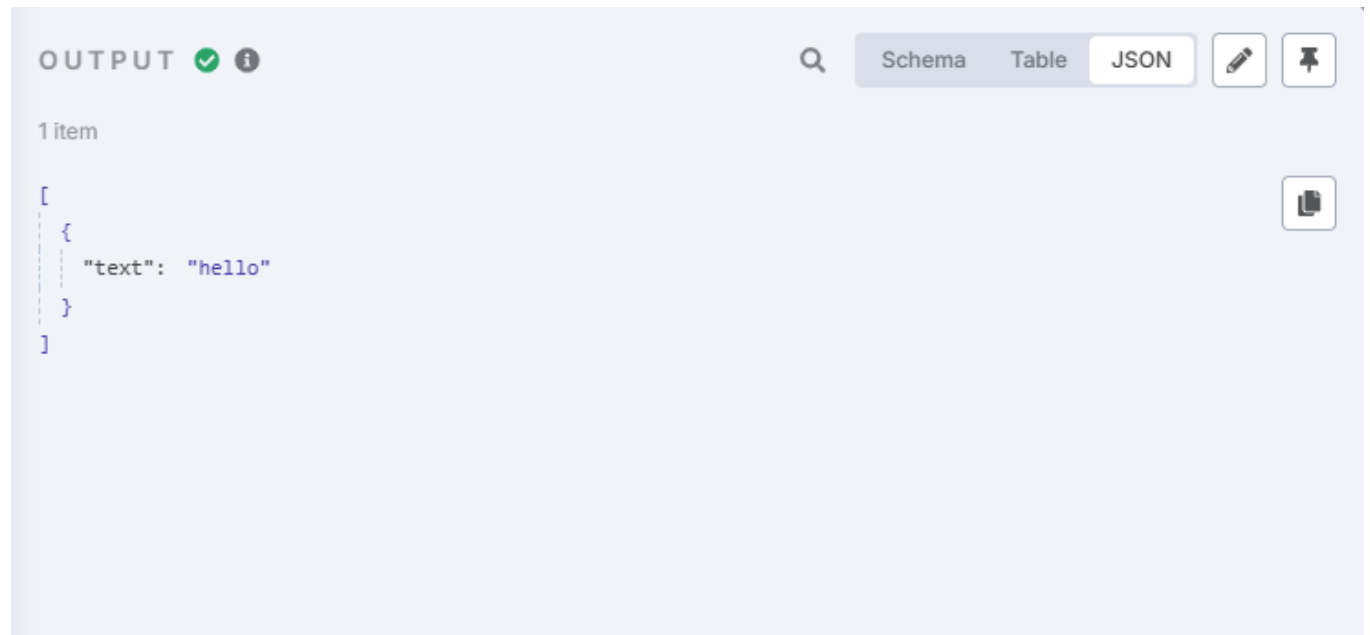
Options

No properties

Add option

Output of text only

1. The text sent by the user is extracted and send to GPT for analysis.



A screenshot of a web-based JSON viewer interface. At the top, the word "OUTPUT" is displayed in a light gray font, followed by a green checkmark icon and a small circular information icon. To the right of "OUTPUT" are three tabs: "Schema", "Table", and "JSON", with the "JSON" tab currently selected and highlighted. Further right are two icons: a pencil (edit) and a pushpin (pin). Below the header, the text "1 item" indicates the number of objects in the array. The main area displays a JSON array containing one object:

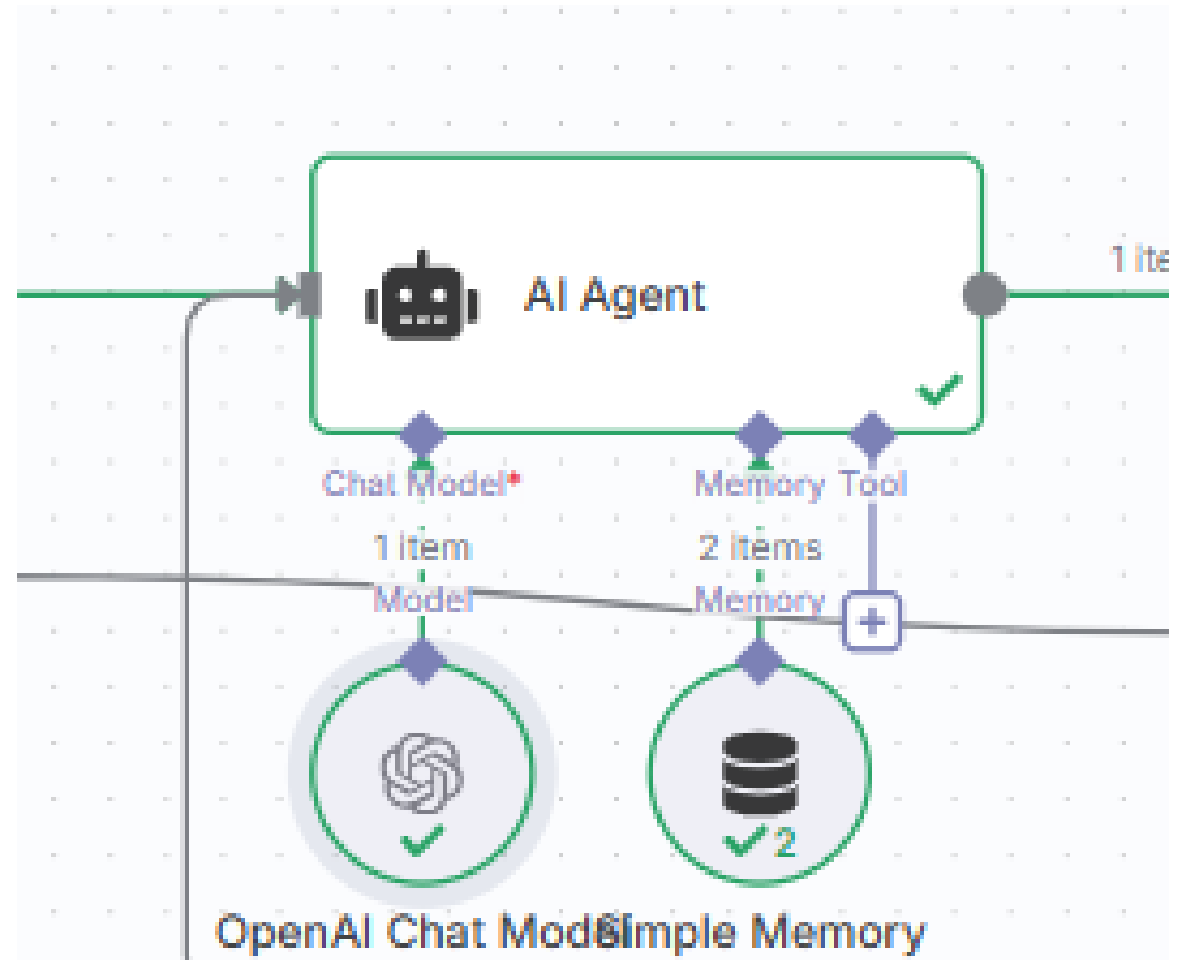
```
[
  {
    "text": "hello"
  }
]
```

 The JSON is syntax-highlighted, with opening and closing brackets in blue and strings in purple. On the far right of the JSON array, there is a small square icon with a document symbol.

AI Agent

In this node:

- GPT-4.0 Mini is prompted with the user message
- Optional system prompt used to control tone/personality
- Temperature and max tokens configured for consistency
- Returned message is passed to the WhatsApp API node



Contents of AI Agent

You are a helpful assistant called Helper AI.
Respond in a friendly tone and act like an assistant.
You are currently talking to `{{ $('WhatsApp Trigger').item.json.contacts[0].profile.name }}`
The current date and time is `{{ $now.toISO() }}`.

 AI Agent

 E

Parameters

Settings



Tip: Get a feel for agents with our quick [tutorial](#) or see an [example](#) of how this node works

Source for Prompt (User Message)

Define below

Prompt (User Message)

fx

`{{ $json.text }}`

hello

Require Specific Output Format



OUTPUT  

 Schema Table JSON

1 item



```
[
  {
    "response": {
      "generations": [
        [
          {
            "text": "Hello Mayank! How can I assist you today?",
            "generationInfo": {
              "prompt": 0,
              "completion": 0,
              "finish_reason": "stop",
              "system_fingerprint": null,
              "model_name": "gpt-4.1-mini-2025-04-14"
            }
          }
        ]
      ]
    },
    "tokenUsage": {
      "completionTokens": 11,
      "promptTokens": 66,
      "totalTokens": 77
    }
  }
]
```

OUTPUT OF GPT 4.0 MINI

Sample input: "Tell me something about diabetes."
Sample output: "Diabetes is a chronic condition that affects how your body processes blood sugar. It can be managed through medication, diet, and exercise."
Output is concise, context-aware, and informative.

WHATSAPP MESSENGER SENDER INPUTS

The final reply is formatted with:

- Message body as GPT output
- Recipient's WhatsApp number
- Optional emoji or signature
- Sent using HTTP POST to the WhatsApp Cloud API endpoint

Parameters

Settings

Docs 

Credential to connect with

WhatsApp respond



Resource

Message



Operation

Send



Sender Phone Number (or ID)

Error fetching options from WhatsApp Business Cloud



Recipient's Phone Number

fx

```
{{ $('WhatsApp Trigger').item.json.messages  
[0].from }}
```



18144188274

Final whatsapp response.

- Asked Helper AI with text " **can you tell me the gas price in Pennsylvania**"
- Helper AI responded with " **Sure, Mayank! As of the most recent data in July 2025, the average cost of regular gasoline in Pennsylvania is approximately \$3.45 per gallon. Keep in mind that prices can vary depending on the city and specific gas station. Would you like information on premium or diesel fuel prices as well?**"

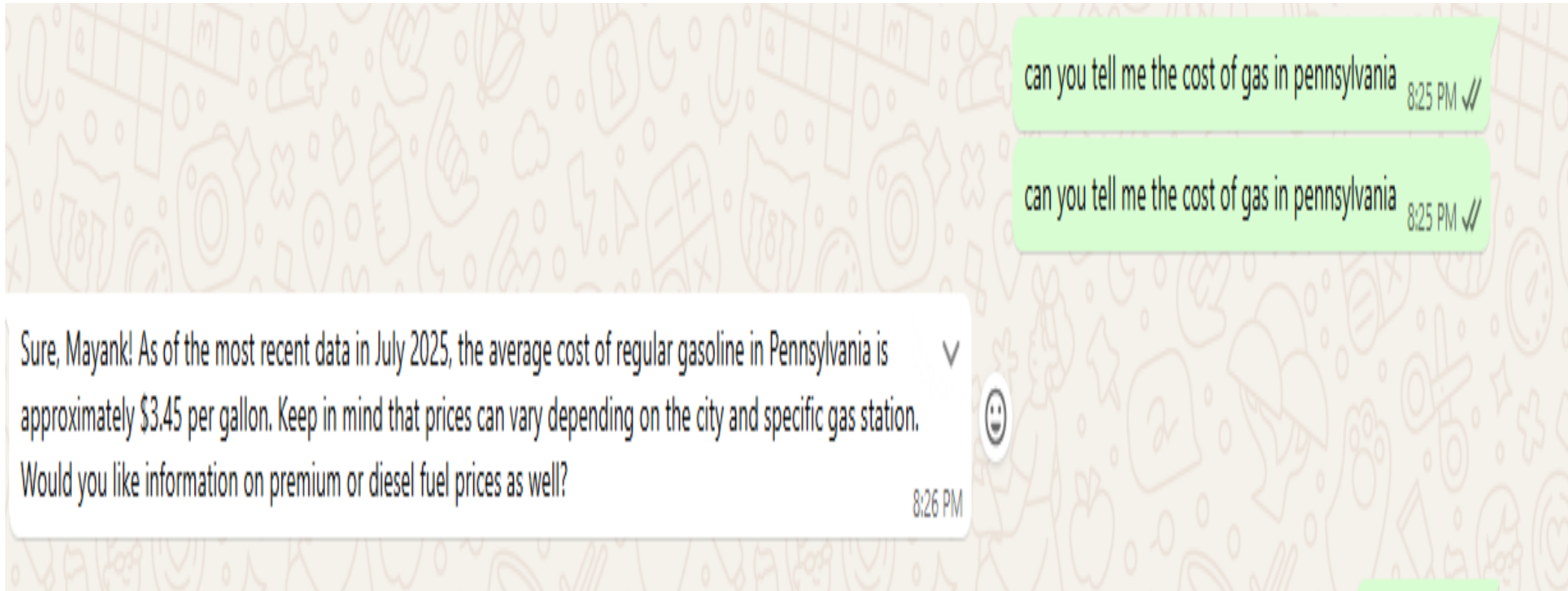


Image input to our AI- Helper

WhatsApp media webhook captures image ID.
n8n downloads the image via WhatsApp API



If the switch detects an Image input it will
redirect it towards the image section and the
whole process will work separately.



Switch will redirect it towards get image from
Whatsapp to .

WHATSAPP GET IMAGE

- Receives image ID from WhatsApp webhook
- Retrieves media URL using WhatsApp Cloud API
- Image is temporarily stored or passed to GPT
- Only JPG and PNG types currently supported

 get image

 Execute step

Parameters

Settings

Docs [↗](#)

Credential to connect with

WhatsApp respond

Resource

Media

Operation

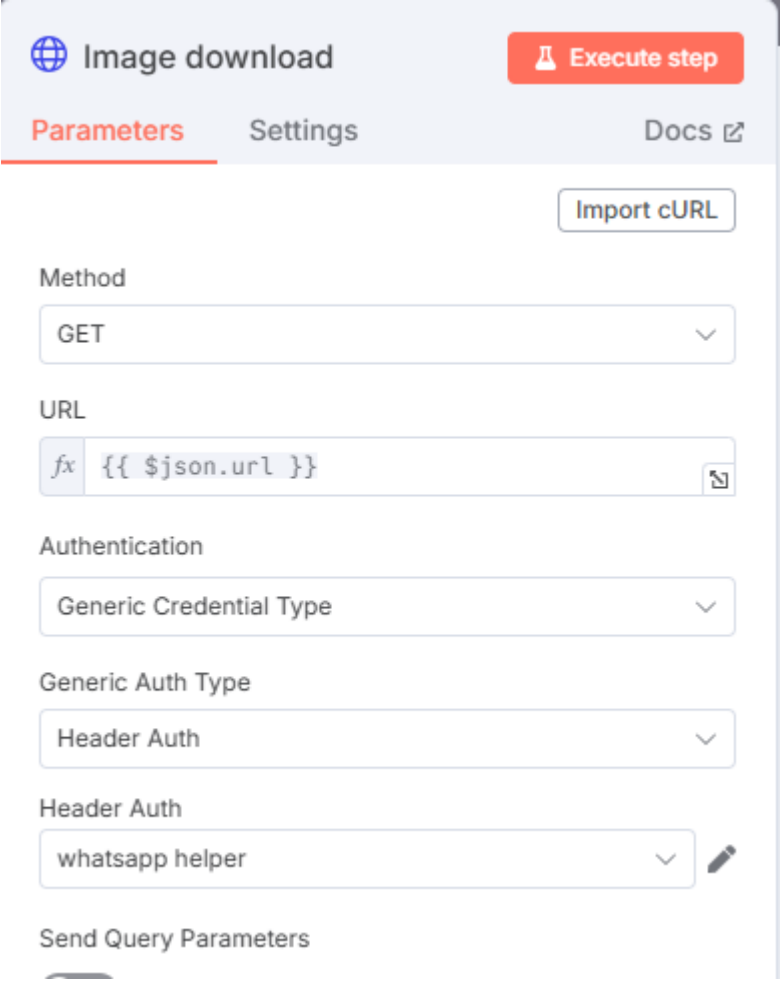
Download

Media ID

fx `{{ $json.messages[0].image.id }}`

IMAGE DOWNLOADER

- Image media ID is converted to a downloadable link
- n8n uses HTTP GET with bearer token
- File is prepared for processing
- Output is sent to the next step: "Analyze Image"



The screenshot shows the configuration interface for the 'Image download' step in n8n. The interface has a light blue header with a globe icon, the step name 'Image download', and an 'Execute step' button. Below the header are tabs for 'Parameters' (active), 'Settings', and 'Docs'. An 'Import cURL' button is located in the top right of the configuration area. The main configuration section includes: a 'Method' dropdown set to 'GET'; a 'URL' field with a JavaScript expression '{{ \$json.url }}'; an 'Authentication' dropdown set to 'Generic Credential Type'; a 'Generic Auth Type' dropdown set to 'Header Auth'; a 'Header Auth' field containing the text 'whatsapp helper'; and a 'Send Query Parameters' section at the bottom.

Image download [Execute step](#)

[Parameters](#) [Settings](#) [Docs](#)

[Import cURL](#)

Method

GET

URL

`fx` `{{ $json.url }}`

Authentication

Generic Credential Type

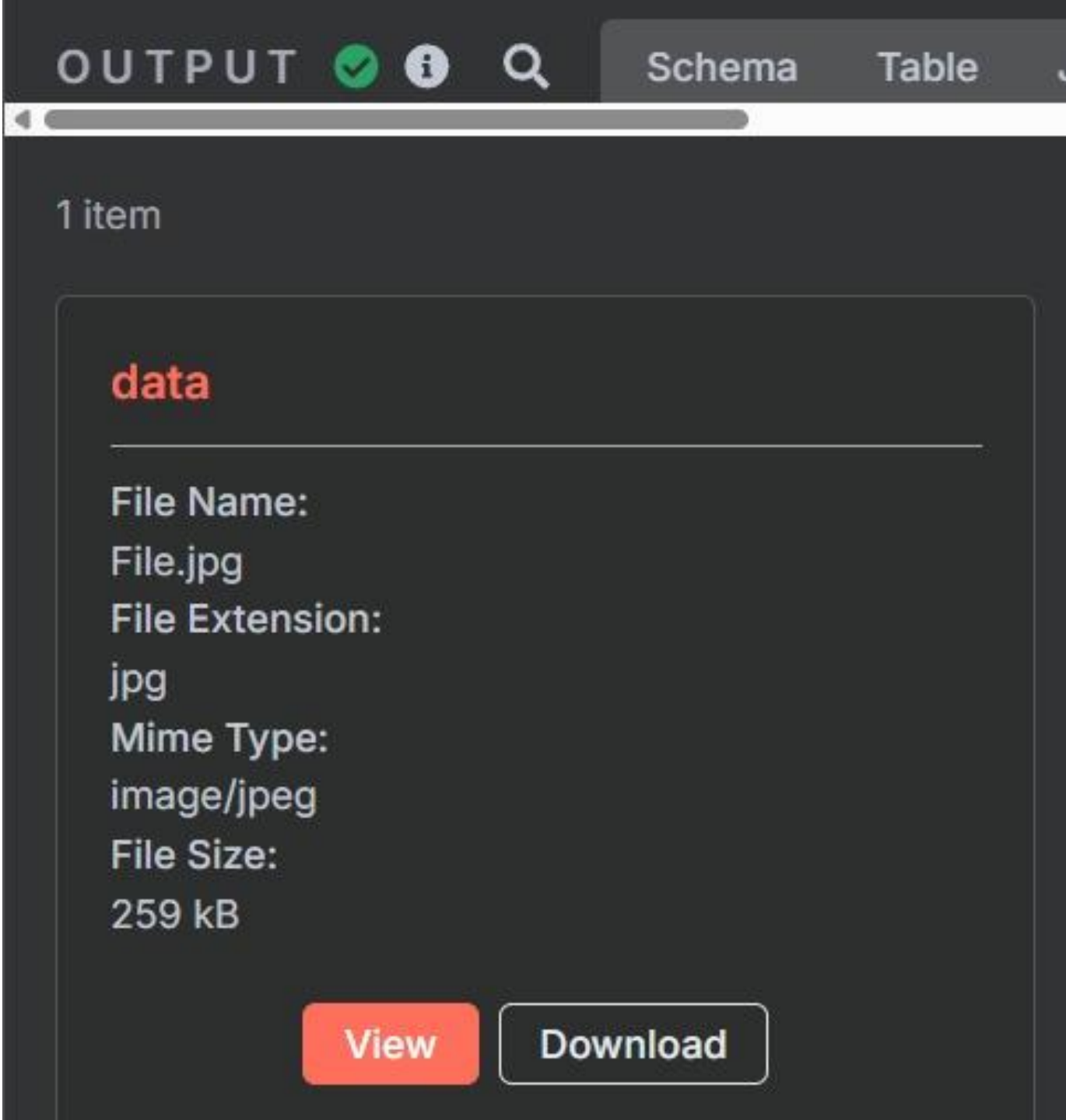
Generic Auth Type

Header Auth

Header Auth

whatsapp helper

Send Query Parameters



Output of Image downloader

The output is a Jpg/image type which is sent to GPT to analyze.

Analyze Image.

- Image URL is embedded in a prompt
- Prompt sent to GPT: *"Describe this image in one sentence."*
- GPT responds with caption or best guess (e.g., "A man hiking in the mountains at sunset.")
- Supports general object detection, context inference

 Analyze image Execute step

Parameters Settings Docs

Credential to connect with

n8n free OpenAI API credits

Resource

Image

Operation

Analyze Image

Model

From list

GPT-4O-MINI

Text Input

Describe this image in detail.

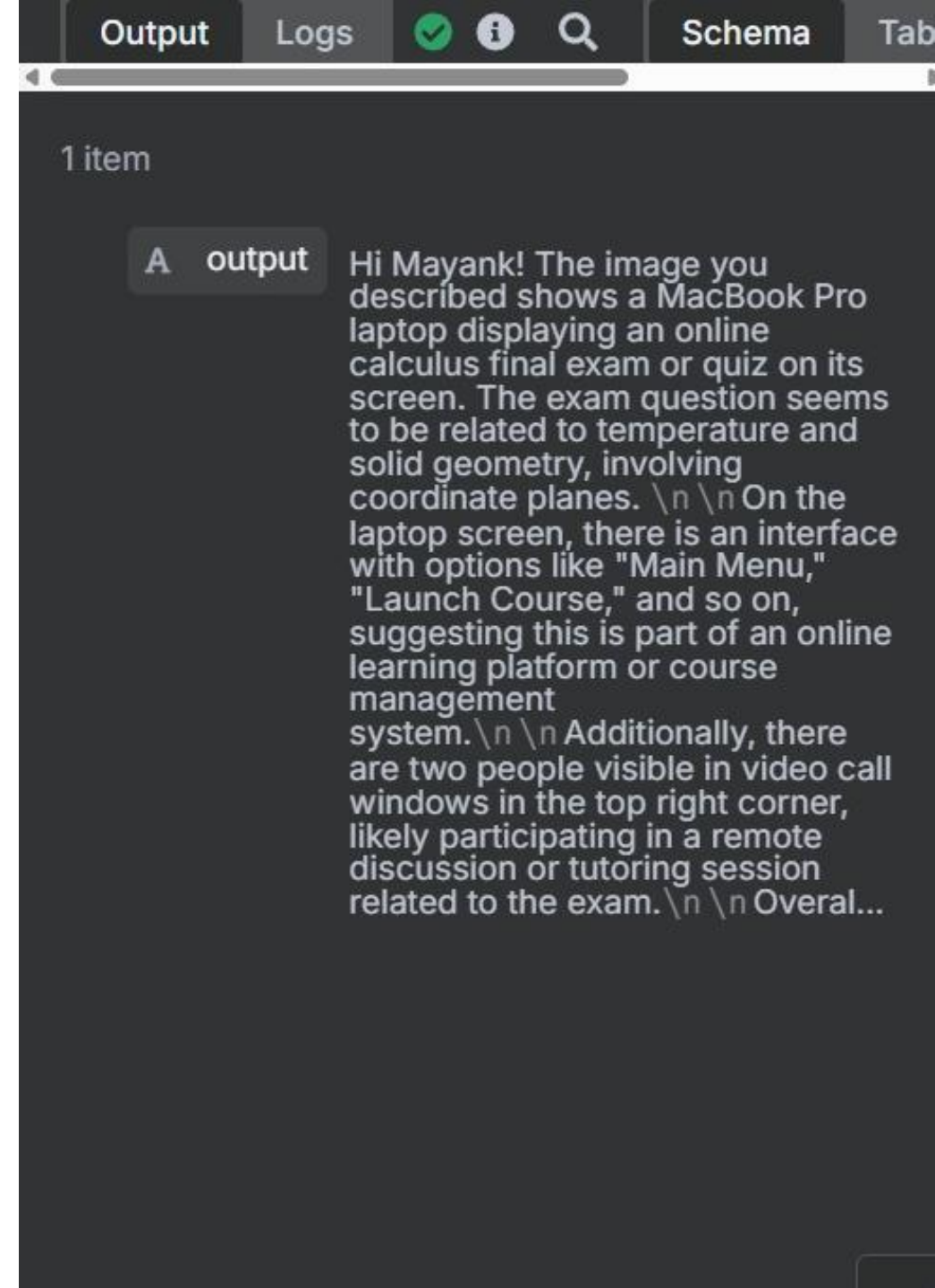
Input Type

Binary File(s)

Input Data Field Name

data

Output of analyze image



Final output on WhatsApp

